Lecture Notes for

Neural Networks I

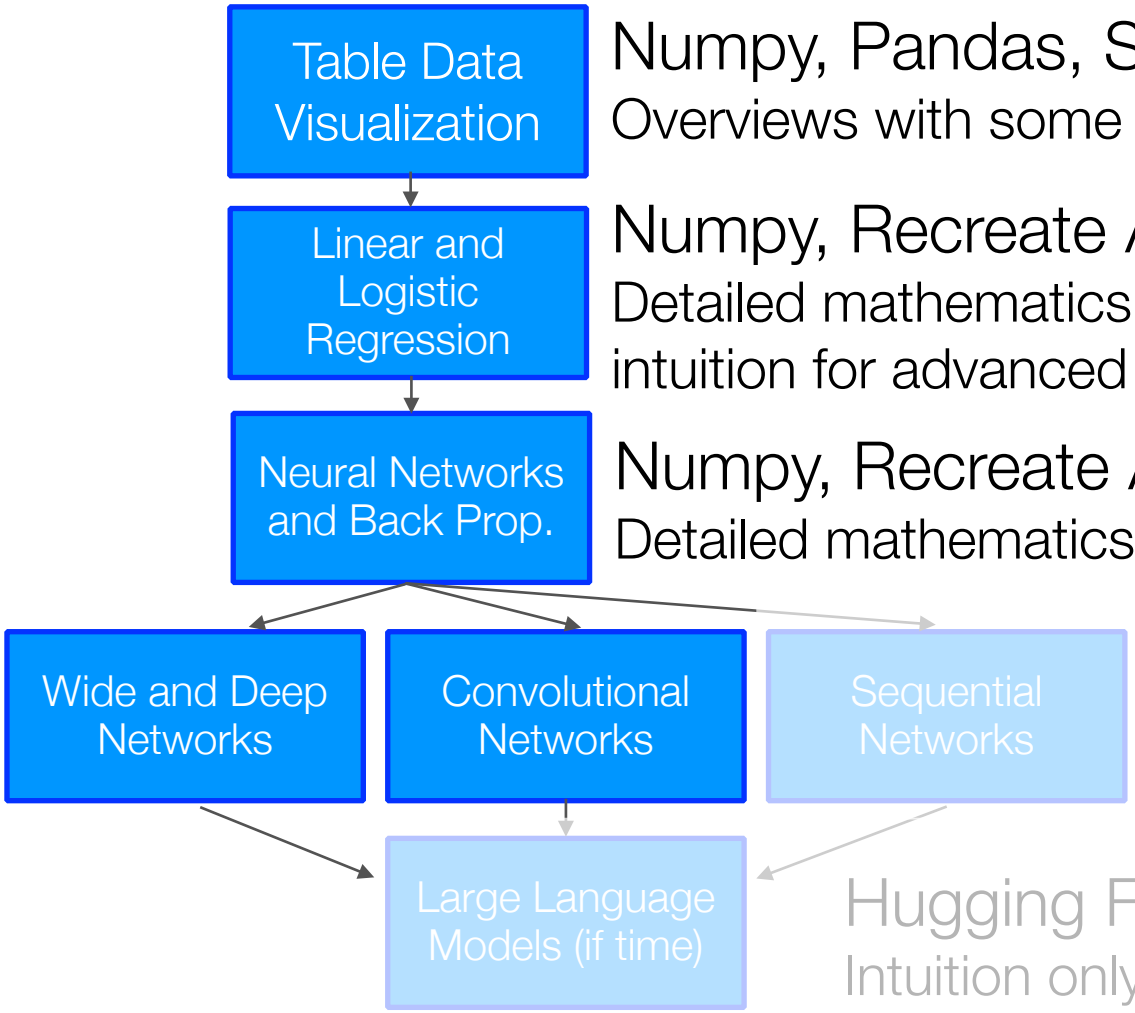


Convolutional Neural Networks

Logistics and Agenda

- Logistics
  - Wide/Deep due soon!
  - Town Hall, if needed
- Agenda
  - Basic CNN Structures
  - CNN Gradient overview

Class Overview, by topic



Numpy, Pandas, Seaborn
Overviews with some in-depth discussion

Numpy, Recreate API for Scikit-learn
Detailed mathematics for simple optimization
intuition for advanced optimization

Numpy, Recreate API for DL Frameworks
Detailed mathematics for NN operations

Hugging Face
Intuition only

Convolutional Neural Networks

STOP making fun of different programming languages

C is FAST

Java is POPULAR

Ruby is COOL

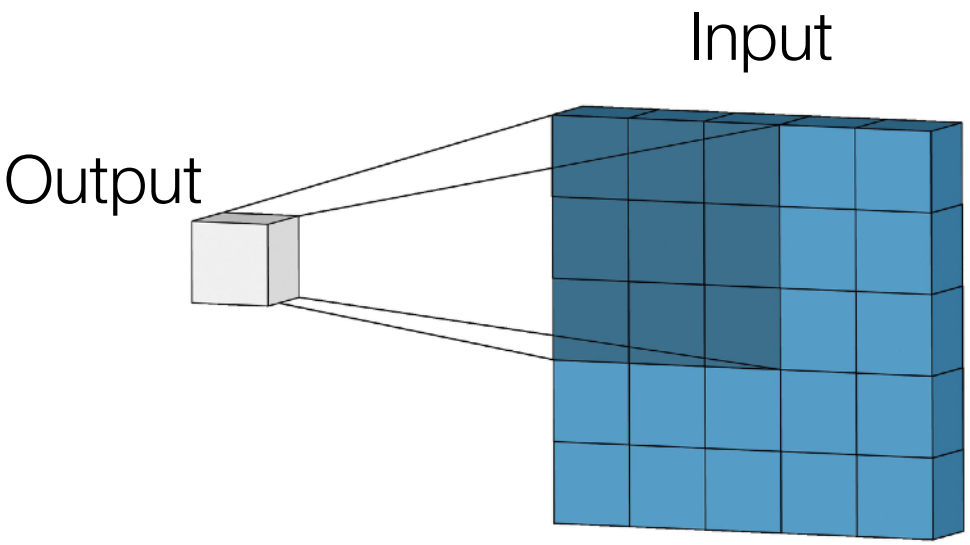
Python is BEAUTIFUL

Javascript

Haskell is INTRIGUING

Extracting Features: Convolution

- For images:
  - kernel (matrix of values)
  - slide kernel across image, pixel by pixel
  - multiply and accumulate



This Example:
3x3 Kernel (dark)
Ignoring edges of input
Input Image is 5x5
Output is then 3x3

https://towardsdatascience.com/intuitively-understanding-convolutions-for-deep-learning-1f6f42faee1

Image Convolution

$$\sum_{a,b \in r,c} \mathbf{I}[i+a, j+b] \cdot \mathbf{k}[a,b] = \sum \left(\mathbf{I}\left[i \pm \frac{r}{2}, j \pm \frac{c}{2} \right] \odot \mathbf{k} \right) = \mathbf{O}[i,j]$$

output image at pixel i,j

input image slice centered in i,j with range r x c

kernel of size, r x c usually $r=c$

0	0	0	0	0	0	0	0	0
0	1	2	3	4	12	9	8	0
0	5	2	3	4	12	9	8	0
0	5	2	1	4	10	9	8	0
0	7	2	1	4	12	7	8	0
0	7	2	1	4	14	9	8	0
0	5	2	3	4	12	7	8	0
0	5	2	1	4	12	9	8	0
0	0	0	0	0	0	0	0	0

input image, $\mathbf{I}$

0	0	0
2	3	4
2	3	4

1	2	1
2	4	2
1	2	1

kernel filter, $\mathbf{k}$ 3x3 r x c

20	21	36	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…
…	…	…	…	…	…	…

output image, $\mathbf{O}$

Convolution Examples

0	0	0
1	0	0
0	0	0

Self test:

What does this do?

A. move left pixel to center

B. move right to center

C. blur



Blur

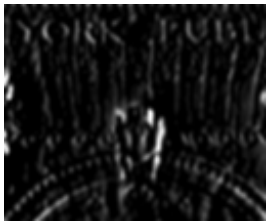
1	1	1
1	1	1
1	1	1





Vertical Edges

-1	0	1
-1	0	1
-1	0	1





Sharpen

0	-1	0
-1	5	-1
0	-1	0





Convolution is linear

with $\alpha = 5$

$$\alpha \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

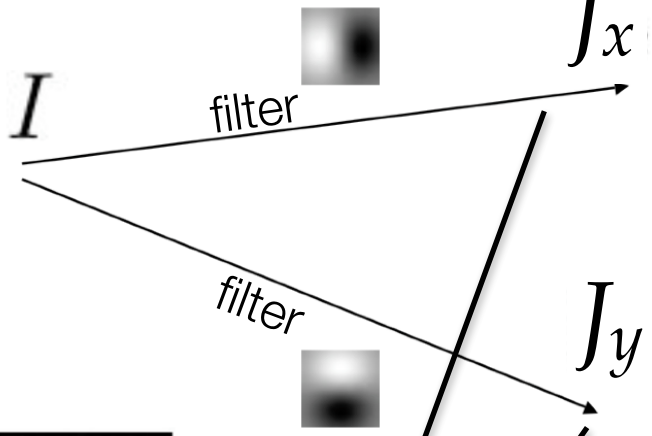
α Image + (Image - Blur) = Sharp

https://setosa.io/ev/image-kernels/

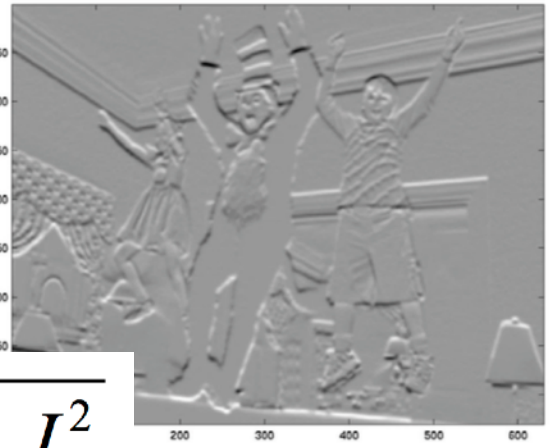
Common operations with convolution

- pixel derivative





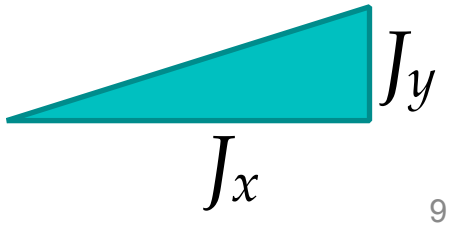






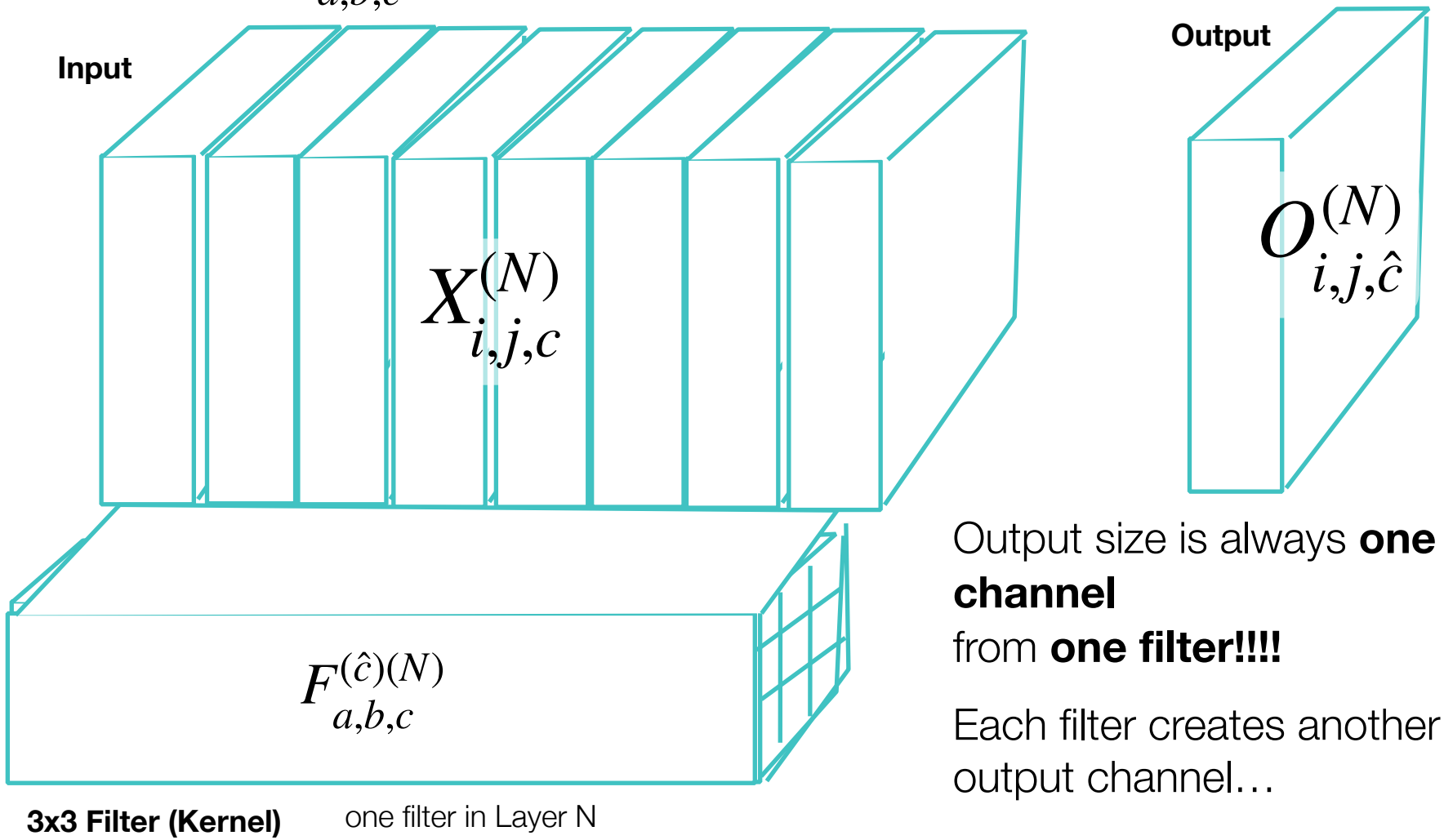
$$\|\nabla J\| = \sqrt{J_x^2 + J_y^2}$$

$$O = \tan^{-1}(J_y / J_x)$$



Convolution in a CNN

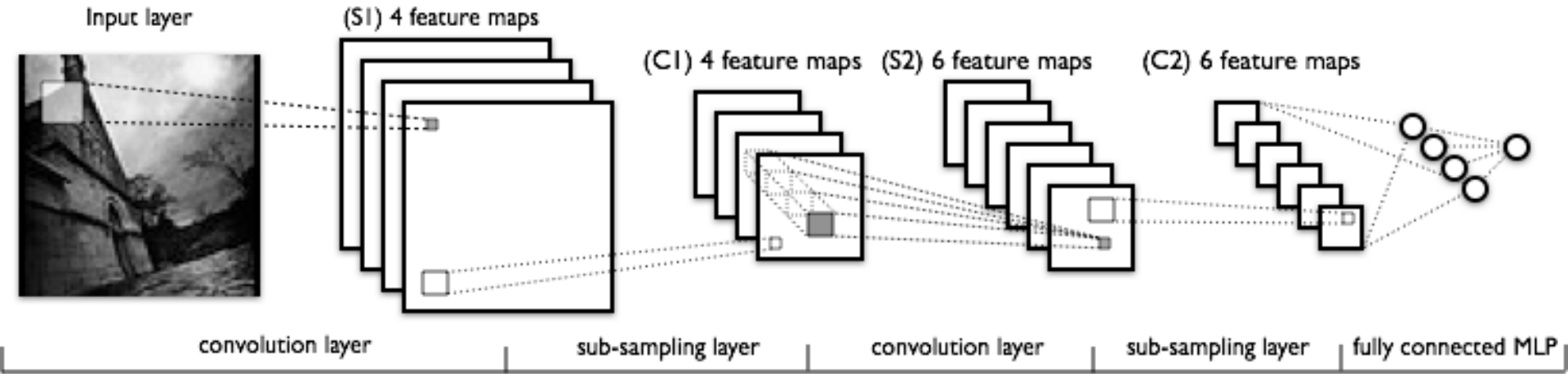
$$O_{i,j,\hat{c}} = \sum_{a,b,c} X[(i+a),(j+b),(k+c)] \cdot F^{(\hat{c})}[a,b,c]$$



Output size is always **one channel** from **one filter!!!!**

Each filter creates another output channel…

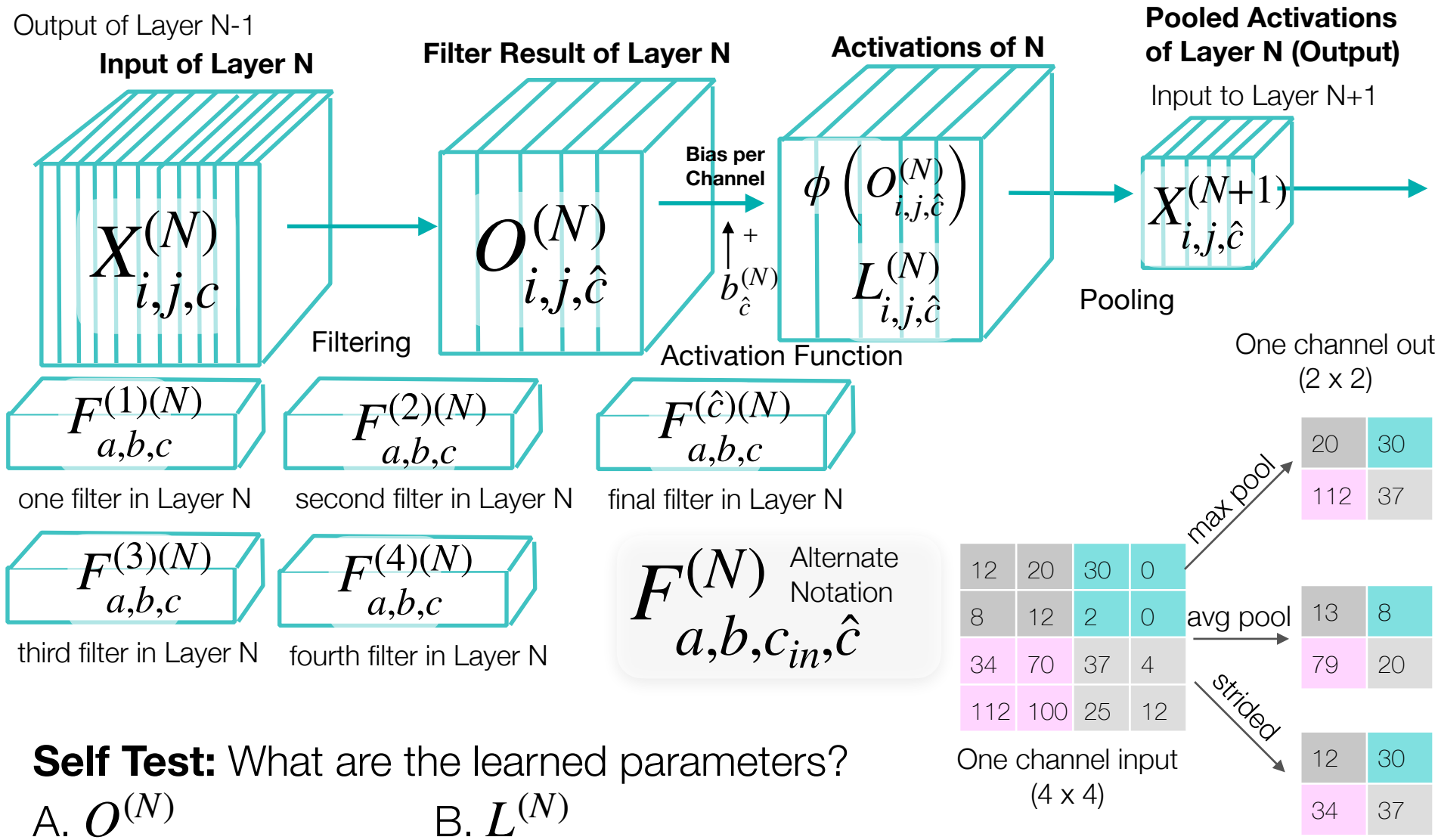
CNN Overview



- **Conv. layer(s):**
  - convolution and activation function (output is activation)
  - pooling downsamples resolution of activation output
    - less dependence on exact pixels
- **Final layers are densely connected**
  - typically multi-layer perceptrons (logistic regression)

Convolutional Layers

Structure of Each Tensor:
Rows x Columns x Channels



Self Test: What are the learned parameters?

A. $O^{(N)}$

B. $L^{(N)}$

C. $F^{(N)}$ and $b^{(N)}$

D. All of these